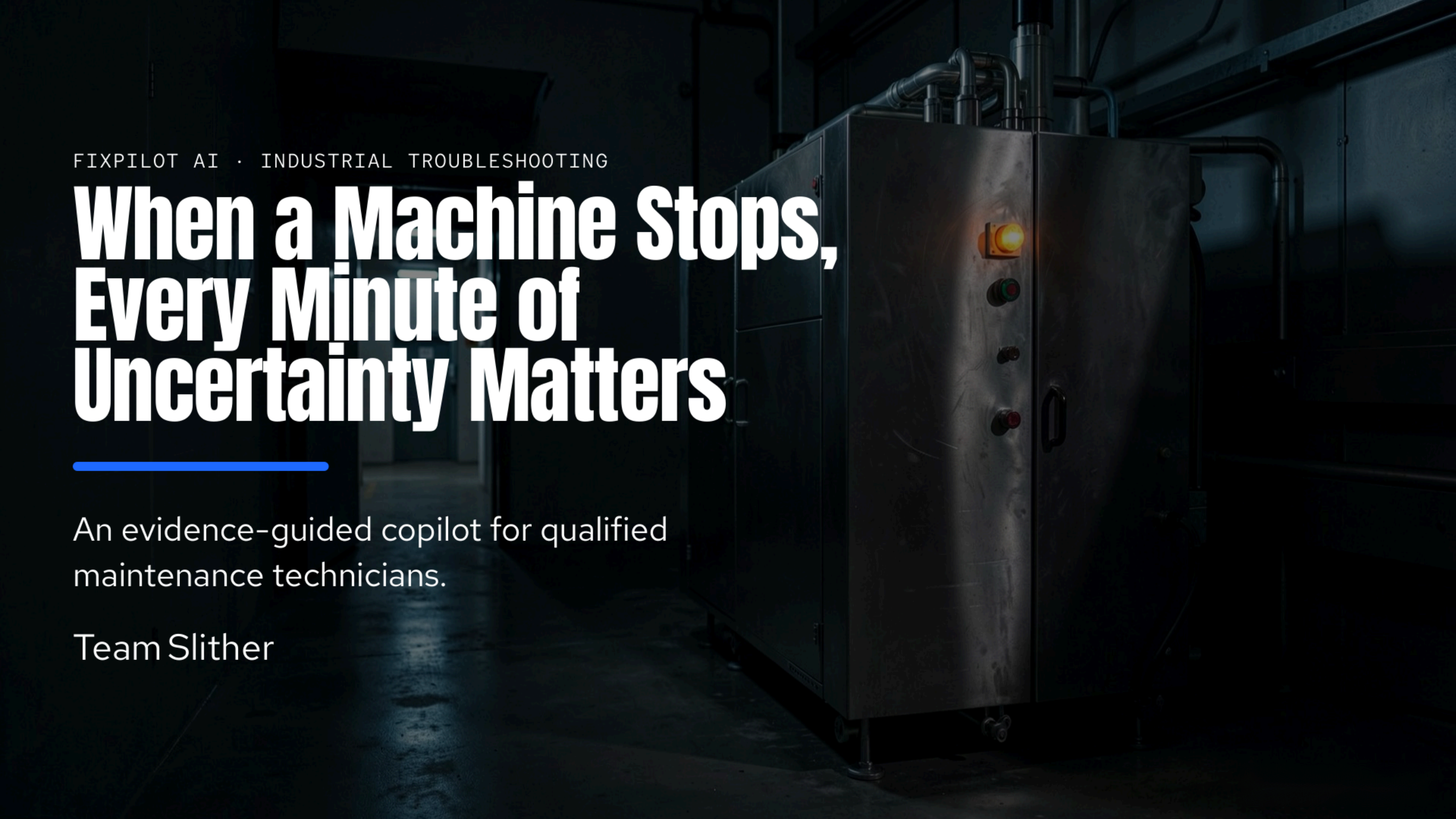


FIXPILOT AI · INDUSTRIAL TROUBLESHOOTING

When a Machine Stops, Every Minute of Uncertainty Matters

An evidence-guided copilot for qualified maintenance technicians.

Team Slither

THE PROBLEM

Technicians lose time stitching together scattered knowledge

Arjun, a maintenance technician at a small or medium-sized plant, faces unfamiliar error codes, long manuals, and little access to experts.

1 · FAULT

Machine stops

2 · SEARCH

Bottleneck: manuals and context

3 · ASK

Wait for an expert

4 · TEST

Try possible causes

5 · DECIDE

Resolve or escalate



The delay is often the search for context.

THE PROPOSED PRODUCT

FixPilot AI turns approved documentation into a troubleshooting workflow

1 · INPUT

Error code, symptoms, or approved manual

2 · RETRIEVE

Relevant passages from equipment docs

3 · DIAGNOSE

Causes and checks; facts vs. hypotheses

4 · ACT OR ESCALATE

Technician reviews, records, escalates

ILLUSTRATIVE EXAMPLE · NOT A PILOT RESULT

Technician in control at the review gate: the assistant proposes, a qualified technician decides.

FIXPILOT UI MOCKUP · SIMULATED FAULT

Fault code · ranked evidence

Possible causes · sensor obstruction · loose connection

Recommended checks · cited manual sections · uncertainty labels

Review gate · facts stay separate from hyp

WHY THIS APPROACH

AI makes the search contextual, explainable, and safer to use

Technician query | **Parse & retrieve** | **Approved evidence** | **LLM reasoning** | **Cited plan** | **Technician review**

TRADITIONAL TOOLS

Keyword search and fixed trees; cross-document reasoning left to the user.

FIXPILOT AI

Natural language, cited evidence, structured plan, flagged uncertainty.

No unsupported guesses as facts

No bypassing safety interlocks

No hazardous procedures

No live machinery controls in MVP

THE BUILD

Three weeks is enough for a focused, testable MVP

WEEK 1 · FOUNDATION

Retrieval works

Deliverable: relevant manual passages for a simulated fault.

WEEK 2 · COPILOT

Evidence-linked answers

Deliverable: end-to-end cited troubleshooting assistant.

WEEK 3 · DEMO

Live demo hardened

Deliverable: functional demo on a simulated machine fault.

SCOPE BOUNDARY

In: simulated faults, approved manuals, cited retrieval. Out: PLC integration, live sensors, digital twins, autonomous control, predictive-maintenance claims.

THE PILOT INVITATION

From Searching Manuals to Evidence-Guided Maintenance

Measure	What we check
Citation support	Answers trace to approved passages
Retrieval correctness	Right passage found for simulated faults
Time to information	Minutes to the next evidence-backed check
Escalation	Unsafe or unsupported prompts flagged

The ask: access to real maintenance workflows and expert judgment.



Validate the workflow where maintenance work actually happens.